Conjugate degradability implies degradability for support-minimal channels with a qubit environment

Abstract

Every finite-dimensional, support-minimal conjugate-degradable channel with a qubit environment is degradable. A bosonic extension lemma converts the conjugate-degrading condition into an ordinary completely positive trace-preserving degrading map. The result includes the regime of input dimension at least four and output dimension larger than the input dimension.

1 Setting and statement

Let A, B, and E be finite complex Hilbert spaces, with $\dim E = 2$, and let $V : A \to B \otimes E$ be an isometry. Define

$$\mathcal{N}(X) = \mathrm{Tr}_E(VXV^*), \qquad \mathcal{N}^c(X) = \mathrm{Tr}_B(VXV^*).$$

Fix the computational basis of E; every transpose below refers to this basis. We use a support-minimal presentation: both $\mathcal{N}(I_A)$ and $\mathcal{N}^c(I_A)$ are positive definite on their displayed output spaces. The construction below uses the full support on B. The condition in the theorem is *conjugate degradability* in the terminology of Brádler, Dutil, Hayden and Muhammad [BDHM10]. On Hermitian matrices, complex conjugation equals transposition; the displayed identity is its complex-linear extension to all input matrices.

Theorem. If a completely positive trace-preserving map $D : \mathcal{L}(B) \to \mathcal{L}(E)$ satisfies

$$D \circ \mathcal{N} = \mathsf{T} \circ \mathcal{N}^c,$$

then a completely positive trace-preserving map $D_0 : \mathcal{L}(B) \to \mathcal{L}(E)$ satisfies $D_0 \circ \mathcal{N} = \mathcal{N}^c$.

The equalities hold on all complex input matrices. In particular, strict conjugate degradability is impossible within this support-minimal environment-qubit class.

Write $d_A = \dim A$ and $d_B = \dim B$. Brádler [B15, Lemma 4] already excludes exclusively conjugate-degradable channels when $d_B \le d_A$. For a qubit environment, the case $d_A \le 3$ also follows from PPT separability in dimensions 2×2 and 2×3 [HHH96]: the complementary Choi matrix is then separable, so the complementary channel is entanglement breaking and its complement is degradable [BDHM10; B15, Section 3.2]. The theorem therefore extends these exclusions to $d_A \ge 4$, $d_B > d_A$, within the stated support-minimal class with a qubit environment.

2 A bosonic extension lemma

Let R be a finite reference space, and let $W \ge 0$ act on $R \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$. Write $W_1 = \mathrm{Tr}_2 W$ and $W_2 = \mathrm{Tr}_1 W$, retaining the reference in both cases.

Lemma. If the 00 reference blocks of W_1 and W_2 are equal, then W_1 has a positive extension $\widetilde{W}$ supported on the symmetric subspace of the two qubits, and both marginals of $\widetilde{W}$ equal W_1.

Proof. Factor $W = ZZ^*$ and split the rows of Z in the order $00, 01, 10, 11$:

$$Z = \begin{pmatrix} A \\ B \\ G \\ D \end{pmatrix}.$$

The equality of the diagonal marginal blocks says $AA^* + BB^* = AA^* + GG^*$, hence $BB^* = GG^*$. There is a unitary U on the column space with $G = BU$. Indeed, the rule carrying each row of B to the corresponding row of G preserves their Gram matrix, hence is an isometry on their spans; extend orthonormal bases of the spans to the common column space. This argument does not require either row factor to be invertible.

Set

$$\widetilde{Z} = \begin{pmatrix} A \\ G \\ G \\ DU \end{pmatrix}, \qquad \widetilde{W} = \widetilde{Z}\widetilde{Z}^*.$$

It is positive and its rows are invariant under swapping the two qubits. Its first marginal has blocks

$$\begin{pmatrix} AA^* + GG^* & AG^* + G(DU)^* \\ GA^* + (DU)G^* & GG^* + (DU)(DU)^* \end{pmatrix} = \begin{pmatrix} AA^* + BB^* & AG^* + BD^* \\ GA^* + DB^* & GG^* + DD^* \end{pmatrix}.$$

This is precisely W_1. Swap invariance gives equality of the two marginals, as required. □

Only diagonal agreement was needed. In particular, the lemma applies when $W_2 = (\mathrm{id}_R \otimes \mathsf{T})(W_1)$.

3 From a purification to a channel

We record the channel reconstruction with its support hypothesis explicit. Let $Z : R \leftarrow B$ have full column rank and let $|Z\rangle = \sum_{r,b} Z_{rb}|r, b\rangle$. If $\sigma \geq 0$ on $R \otimes E$ satisfies $\mathrm{Tr}_E\, \sigma = ZZ^*$, then there is a channel $D_0 : \mathcal{L}(B) \to \mathcal{L}(E)$ such that

$$(\mathrm{id}_R \otimes D_0)(|Z\rangle\langle Z|) = \sigma.$$

To see this, put $L = (Z^*Z)^{-1}Z^*$, so $LZ = I_B$, and define

$$J = (L \otimes I_E)\sigma(L^* \otimes I_E).$$

Then $J \geq 0$ and $\mathrm{Tr}_E\, J = I_B$. Thus J is the unnormalized Choi matrix of a channel: a positive factorization of J, with each column reshaped into an E-by-B matrix, gives Kraus operators whose adjoint products sum to I_B. The marginal identity also implies that σ is supported on $\mathrm{ran}\, Z \otimes E$. Indeed a vector in the complementary subspace has zero quadratic form against the positive matrix σ and therefore lies in its kernel. Since ZL is the orthogonal projection onto $\mathrm{ran}\, Z$, applying $Z \otimes I_E$ to J recovers σ. This proves the displayed identity.

4 Proof of the theorem

Take an unnormalized maximally entangled vector on a reference copy of A and A, and apply V to its second factor. Reorder the resulting purification into reference/environment/output order. Denote it by $|Z\rangle$, where now $Z : (A \otimes E) \leftarrow B$. Its first marginal is the unnormalized Choi matrix $\rho = J(\mathcal{N}^c) = ZZ^*$. Full output support makes Z full column rank: Z^*Z is the transpose of $\mathcal{N}(I_A)$.

Apply the assumed channel D to the output factor. The resulting positive matrix W on $A \otimes E \otimes E$ has first marginal ρ, by trace preservation, and second marginal

$$J(D \circ \mathcal{N}) = (\mathrm{id}_A \otimes \mathsf{T})(\rho).$$

The extension lemma gives a bosonic $\widetilde{W}$ with both marginals equal to ρ. The reconstruction argument gives a channel D_0, defined on the whole output algebra, which maps $|Z\rangle\langle Z|$ to $\widetilde{W}$. Taking its second qubit marginal yields $J(D_0 \circ \mathcal{N}) = \rho = J(\mathcal{N}^c)$. Injectivity of the Choi correspondence proves the theorem. □

5 References

- [**BDHM10**] K. Brádler, N. Dutil, P. Hayden and A. Muhammad, *Conjugate degradability and the quantum capacity of cloning channels*, Journal of Mathematical Physics **51**, 072201 (2010), DOI, arXiv:0909.3297.
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